AISHWARYA VINOD MENON

+1 (214) 836-5055 | aishwaryamenon.v@gmail.com | Linkedin | Github

OBJECTIVE

Data Engineer with 2+ years of experience, seeking opportunities in data science where I can combine a strong foundation in data pipelines, data qualit y, and large-scale processing with machine learning, statistical modeling, and experimentation to generate meaningful insights and support data-driven decisions.

SKILLS

- **Programming, Data Engineering & Cloud**: Python, PySpark, SQL (MySQL, PostgreSQL, SQL Server), Java, Azure (Data Factory, Databricks, Data Lake, Storage, Synapse), Apache Kafka, dbt, ETL/ELT, Data Modeling, Workflow Orchestration, Data Pipeline Development, Data Quality Checks
- **Databases, Warehousing & BI**: Snowflake, SQL Server, PostgreSQL, Power BI, Data Visualization, Data Governance, Data Warehousing Solutions
- **Machine Learning / Data Science**: Predictive modeling, feature engineering, time-series forecasting, supervised and unsupervised learning, healthcare modeling, Model Evaluation, ML Pipeline Deployment, Git, GitHub Actions
- **Certifications:** Databricks certified Data Engineer Associate

EDUCATION

The University of Texas at Dallas
Master of Science, Computer Science **GPA:** 3.8/4.0
Coursework: Big Data Management & Analytics, Database Management, Machine Learning, Artificial Intelligence, Natural Language Processing, Data Structures & Algorithms

R.M.D. Engineering College
Bachelor of Technology, Computer Science **GPA:** 3.9/4.0

WORK EXPERIENCE

Indrasol **Jan 2025 - Nov 2025**
Associate Data Engineer
- Contributed to the development of a professional content recommendation system using collaborative filtering and pre-trained Sentence-BERT embeddings, improving relevance of recommendations for end users.
- Implemented ETL scripts on Azure Databricks to clean, transform, and integrate structured and semi-structured datasets, providing high-quality inputs for model training.
- Assisted in feature engineering and similarity scoring, combining embeddings with collaborative filtering outputs to improve recommendation accuracy.
- Created dashboards to visualize recommendation performance and key insights for stakeholders, enabling iterative tuning of model parameters.
- Worked closely with cross-functional teams to standardize data processing workflows and integrate recommendation outputs into internal systems.

Danala Analytics **May 2021 - Jun 2023**
Data Engineer
- Engineered and maintained scalable end-to-end data pipelines using PySpark, SQL, and Databricks, improving data delivery speed by 45% and providing clean, structured datasets for ML and predictive modeling.
- Automated ETL/ELT workflows to integrate over 15 TB of structured and unstructured data from EHR systems (Epic/Cerner), claims, and patient engagement sources into Azure Data Lake, Snowflake, and Synapse, reducing load times from 6 hrs to 2.5 hrs with 99% uptime.
- Collaborated with data scientists to prepare analytical datasets, accelerating predictive modeling for patient outcomes and readmission by 35% while ensuring performance and usability.
- Designed and optimized data models (Star and Snowflake schemas) using dbt and Databricks SQL, improving reporting efficiency, reducing dashboard latency by 60%, and structuring data for ML feature engineering.
- Built metadata-driven ETL frameworks and automated workflows with alerting and retry logic, reducing manual intervention by 70% and onboarding time for new data sources by 50%, enabling rapid data science experiments.
- Ensured HIPAA-compliant data governance with encryption, anonymization, and role-based access control, maintaining secure and audit-ready datasets for ML and analytics.
- Conducted data quality assessments with Great Expectations, achieving 98% accuracy across ingestion layers to provide reliable, high-quality datasets for modeling.
- Integrated FHIR APIs and HL7 feeds for real-time data collection, supporting operational analytics and continuous ML model inputs.
- Delivered interactive Power BI and Tableau dashboards, reducing reporting time from days to hours and providing actionable insights that guided data-driven decision-making.
- Worked in Agile sprints with cross-functional teams to deploy production-grade pipelines, track lineage, and maintain CI/CD workflows, ensuring scalable and reproducible datasets for ML projects.

PROJECTS

Healthcare Readmission Prediction
- Developed an end-to-end data pipeline using the MIMIC dataset to predict 30-day hospital readmissions. Built ML models and automated workflows for scalable, insight-driven healthcare analytics on Azure.

Skiing Search Engine
- Built a skiing search engine using Apache Nutch and Solr for indexing and retrieval, and applying query expansion, tokenization, stemming, and K-M eans clustering to improve result relevance, with a Python backend and lightweight front-end for interactive search.